5300 PROJECT PROPOSAL

Topic Name: Using Animations to Teach the Foundations of Network Security

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Fall 2026

Individual project (no partner)

PROJECT DESCRIPTION

My project fits into the **visualization** category: a website of short animated, interactive lessons (modules) that teach network security foundations to anyone! The modules build from basic understanding to more complex networking topics.

The website that I plan to develop will use a single HTML/JavaScript/SVG file hosted free on my GitHub page repository. Every module will have interactive buttons, so learners move through the animation at their own pace. There will also be a short 5-question self-check quiz at the end of each module.

I'll be using the animation code that I built for my Creative Component Presentation for CPRE 5990 as a template to build off of for this project.

Below is a link to my Creative Component Presentation as a reference point to show the style of animation that I plan to implement: https://captainrussell.github.io/Creative-Component/slides/#6

TARGET AUDIENCE

Anyone curious about how the internet protects their information: high school students, parents, and the general public. The modules will be written at a high school level. I also have the additional duty of Cybersecurity Liaison for the Air Force ROTC Department at Iowa State, so it could be a fun lesson for my cadets as well!

AUDIENCE SKILL LEVEL

No computer engineering, networking, or cyber background is assumed; learners only need a web browser and basic computer and arithmetic skills. Technical ideas like packets, encryption keys, hashing, and firewalls are taught in the module that needs them, using everyday comparisons (postcards, paint mixing, fingerprints, a bouncer at the door) to avoid using any jargon or complex math/terminology. As an educator myself as an assistant professor, my goal when building lesson plans is to make it as digestible and understandable as possible for my students. I would much rather teach to a level where everyone can understand, then to teach at a much higher level where only a few of my students can grasp the concepts effectively.

LEARNING OBJECTIVES

After a participant views or interacts with the project, they should be able to ...

Objective 1 – Describe how a message is split into packets and passed from router to router.
Objective 2 – Explain how two people can safely agree on a shared secret while someone watches everything they send.
Objective 3 – Explain why websites store a password's fingerprint (hash) instead of the password itself, and why a long passphrase takes far longer to guess than a short password.
Objective 4 – Describe how a firewall decides which traffic to let in, and why a flood of traffic from many machines (DDoS) can knock a website offline even when nothing is stolen.

Each objective is met when a learner scores 80% (4 of 5) or higher on that module's quiz.

RELEVANCE OF TOPIC

Module 1 teaches packets and routing as postcards passed hand to hand across a map of routers.

Module 2 covers key exchange (Diffie-Hellman) and the steps that lets HTTPS build a shared secret over a network that someone is watching.

Module 3 covers how servers store password hashes and how guessing attacks work against them.

Module 4 covers firewall rules, ports, and denial of service.

RESEARCH AND SUPPORTING MATERIALS

For my CPRE 5990 creative component with Dr. Daniels, I built animations for a card stream cipher. The new research is getting the networking and security details right and explaining them simply: Kurose and Ross [1] for packets, routing, Wi-Fi, and firewalls, Diffie and Hellman [2] and the TLS 1.3 standard [3] for key exchange and HTTPS in Module 2, Morris and Thompson [4] and NIST SP 800-63B [5] for password storage and guessing attacks in Module 3, and NIST SP 800-41 [6] and Mirkovic and Reiher [7] for firewall rules and DDoS in Module 4.

REQUIRED RESOURCES

To build it:

- A laptop, code editor, Git, and my free GitHub account.
- Plain HTML, JavaScript, and SVG (no frameworks, server, or database).
- My CPRE 5990 animation code as the template's starting point.
- ~5 testers with no cyber background (friends, family, or cadets in non-technical majors to test it on).

To use it: a modern web browser. All graphics are open source or generated SVG drawings.
(Use of a LLM for SVG drawings may be used to increase development time, and will be appropriately annotated when used for attribution and adherence to the student code of conduct)

REFERENCES (IEEE STYLE)

[1] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed. Hoboken, NJ, USA: Pearson, 2021.

[2] W. Diffie and M. E. Hellman, "New directions in cryptography," *IEEE Trans. Inf. Theory*, vol. IT-22, no. 6, pp. 644–654, Nov. 1976.

[3] E. Rescorla, "The Transport Layer Security (TLS) Protocol Version 1.3," IETF, RFC 8446, Aug. 2018.

[4] R. Morris and K. Thompson, "Password security: A case history," *Commun. ACM*, vol. 22, no. 11, pp. 594–597, Nov. 1979.

[5] P. A. Grassi et al., "Digital identity guidelines: Authentication and lifecycle management," Natl. Inst. Stand. Technol., Gaithersburg, MD, USA, NIST SP 800-63B, Jun. 2017.

[6] K. Scarfone and P. Hoffman, "Guidelines on firewalls and firewall policy," Natl. Inst. Stand. Technol., Gaithersburg, MD, USA, NIST SP 800-41 Rev. 1, Sep. 2009.

[7] J. Mirkovic and P. Reiher, "A taxonomy of DDoS attack and DDoS defense mechanisms," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 34, no. 2, pp. 39–53, Apr. 2004.

PROJECT TIMELINE

Dates	Task	Needs first	Done when
Oct 12–18	Convert my CPRE 5990 animation website into a reusable template	Nothing	A blank module with navigation options and a clear GUI
Oct 12–25	Write scripts and storyboards for all four modules	Nothing	Every module has a high-level story and script
Oct 19–Nov 8	Build Modules 1–2	Template	Modules 1 and 2 run end to end; R&C Dev submitted Nov 8
Nov 9–25	Build Modules 3–4 and write quiz questions	Module 1's packet animation	Modules 3 and 4 run end to end; every module has 5 quiz questions
Nov 26–29	Thanksgiving catch-up time	Nothing	Anything that fell behind is finished
Nov 30–Dec 2	Test with family/cadets for clarity and flow, write the teaching guide	All modules	Feedback and before-and-after quiz scores are recorded
Dec 3–5	Implement feedback and run final checks	Test run	Fixes are live and the project is ready for submission

If I fall behind: I plan to shrink the scope of the 4th module to not include the DDOS section.

Task Dependencies & Potential Issues

Every module depends on the template, and Modules 2 and 4 reuse Module 1's packet animation (the coffee-shop payoff and the traffic at the firewall door). Possible slowdowns:

- **The template.** My CPRE 5990 slides have the baseline for the animation code/style which will be re-used and built upon.
- **Writing for a general audience.** Simple without being wrong is hard. A non-technical reader will check each script before I animate it.
- **Other classes.** My CYBSC 5360 Project II and Term Paper will likely overlap Modules 1 and 2 which will require some deconfliction of my time and priorities.

Expected Outcomes, Evaluation, and Assessment

What I'll turn in:

- A website with four animated modules, one for each learning objective (see the table below).

#	Module	What the learner sees and does
1	Postcards on the internet	A message is chopped into packets and passed hand to hand across a map of routers, readable like a postcard by anyone holding it.
2	Mixing secret colors	Two people each add a private paint color to a shared public one, swap the mixtures, and end up with the same secret color.
3	Crack the password	A "blender" shows how changing one letter completely changes a password's fingerprint (hash), which is what websites store instead of the password.
4	Guard the gate	Packets line up at the network's door, and learners set simple bouncer (firewall) rules, scored on attacks blocked versus real visitors turned away. (Time permitting: Also have a DDoS visualization)

How I'll know it works:

- **It's correct.** Packets and routing match [1], the key exchange matches [2] and [3], password storage matches [4] and [5], the firewall and flood rounds match [6] and [7], and the crack-time counter's math checks out by hand.
- **It teaches.** 3 to 5 people with no cyber background take a short quiz before and after using the modules. I'm looking for scores to go up, not just for people to finish.
- **It hits the objectives.** Testers score 80% or higher on each module's quiz. (at least, they should perform better from the before/after test)

Personal Learning Goals

In my creative component I studied a stream cipher on its own. Here I want to step back and see how security works across a whole network: what is exposed as data travels, how strangers agree on a secret key, how passwords are stored and attacked, and how networks defend their front door. Even more, I want to get better at explaining technical ideas to people with no background, a skill I use daily teaching AFROTC cadets and will need as a future instructor in the cyber field. I also hope to have allot of fun with this project!